

CS6042 – INTERACTIVE MEDIA PROJECT/WORKSHOP-2



UNIVERSITY OF
LIMERICK
OLLSCOIL LUIMNIGH

Modular & Interactive Sculpture

PROJECT TITLE : *THE COSMOS- A JOURNEY THROUGH
CONNECTION AND COLLAPSE*

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PURPOSE OF THE PROJECT

The main goal of this project was to explore what interactivity actually looks like when it lives inside a physical object rather than a screen or a button. The brief pushed us to build a sculpture that doesn't just react to one action, but responds to the ongoing relationship between its different parts and to make that feel meaningful to whoever engages with it.

We wanted something that anyone could pick up and interact with straight away, but that also had enough depth to keep people curious. If someone just places a single cube and watches what happens, they're already part of the piece. If someone connects all four and then deliberately introduces the Black Hole knowing what it does they've essentially told a small story with their hands. We wanted people to feel like they were making something, not just pressing a button.

On a more practical level, the project was also a way for us to work through the entire pipeline of interactive media from start to finish: physical making, electronics, data communication, and real-time audiovisual software. Every stage threw up its own problems, and solving them pushed us to build skills across a lot of different areas fabrication, coding, design, and performance and then pull it all together into one system that actually worked.

WHY THE SPACE THEME?

Choosing a cosmic theme felt personal before it felt like a concept. Space, stars, and the formation of the universe have always been fascinating, the stories humans tell about the cosmos, from ancient myths to modern science, are some of the oldest ways we've made sense of the world. The universe is the one thing every culture, every era, and every person has in common. That felt like the right backdrop for a piece about connection and collapse.

It also made practical sense. A solar system builds up one piece at a time, and every new body changes the gravitational pull on everything else. That maps almost perfectly onto how our resistor network works each new cuboid changes the circuit, shifts the voltage, and alters the whole system. At an abstract level, the physics of our sculpture and the physics of a solar system are telling the same story.

The theme also gave each cuboid a clear visual and sonic identity. The star, Earth, Moon, and Black Hole are immediately recognisable to anyone. They already carry emotional weight warmth, life, mystery, destruction and we could lean into that through TouchDesigner and sound/the music without having to spell it out. The audience already knows what these things mean. Our sculpture just puts that understanding into their hands.

ARTIST STATEMENT

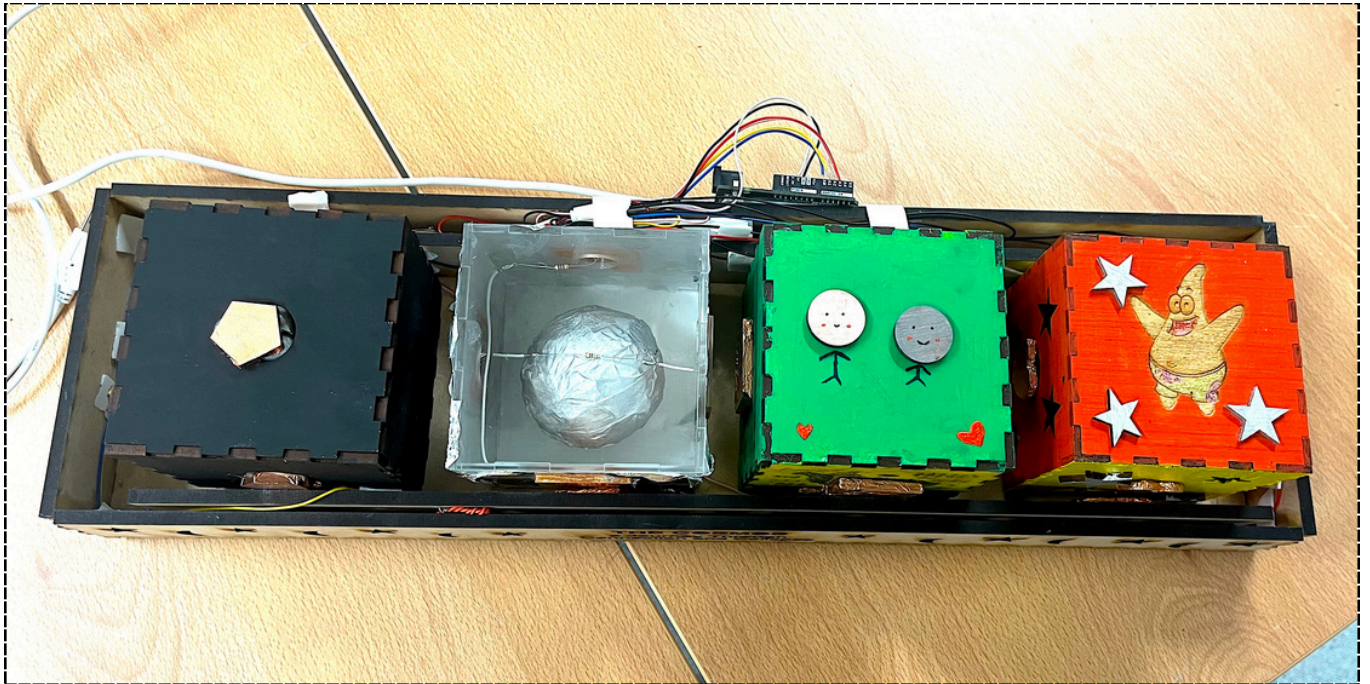
The Cosmos is an interactive sculpture that invites people to help build and ultimately destroy a solar system. It taps into something ancient: that urge to look up at the night sky and search for meaning. But here, that meaning isn't given to you. You make it yourself, one piece at a time.

Inside a box in a quiet room, four laser-cut cuboids wait to be discovered. Each one stands for something: the Star, the Earth, the Moon, and the Black Hole. On their own, they already feel alive. The Star glows with heat. The Earth seems to breathe. The Moon sits in its own stillness. The Black Hole just waits. Each one feels whole and yet somehow unfinished.

When you start connecting them, things begin to change. The Star and Earth together fill the projection with warmth and life. Bring in the Moon, and something clicks three bodies in balance, the system at its most complete. A single piece of music carries through the whole experience, steady and unbroken. But every time a new cuboid is placed, the music dips for a moment, like the universe pausing to adjust. Then it comes back. Until the Black Hole is placed. The moment that fourth cuboid connects, a whirlpool swallows the screen. The light fades. Everything goes dark.

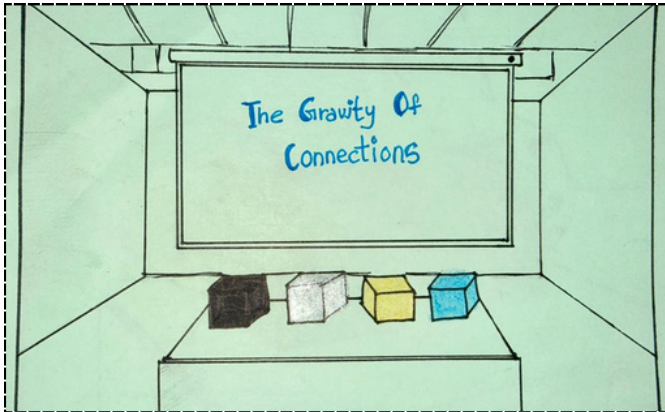
This piece treats interactivity as something more than entertainment it's about responsibility. You don't just watch the cosmos fall apart. You're the one who ends it. Every choice you make carries real weight within the work.

It leaves you with one quiet question: **what does it feel like to hold a universe in your hands, knowing that one more move will destroy it?**



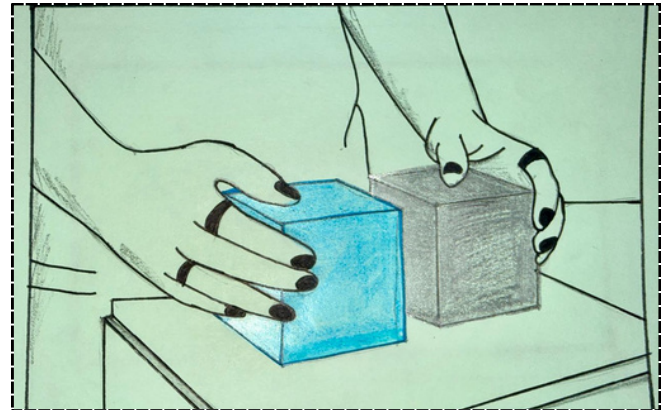
STORYBOARD & NARRATIVE FLOW

Early Stage Storyboard Page



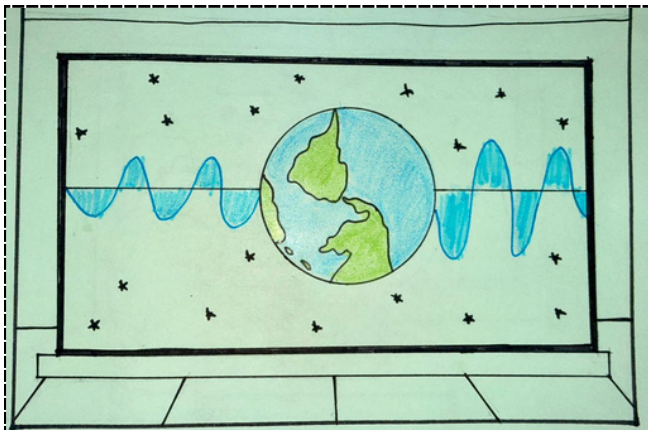
Frame 1: The Void

- Action: Silent space, no interaction
- Description: A room, with 4 cubes present on the table and a projector in-front of it.



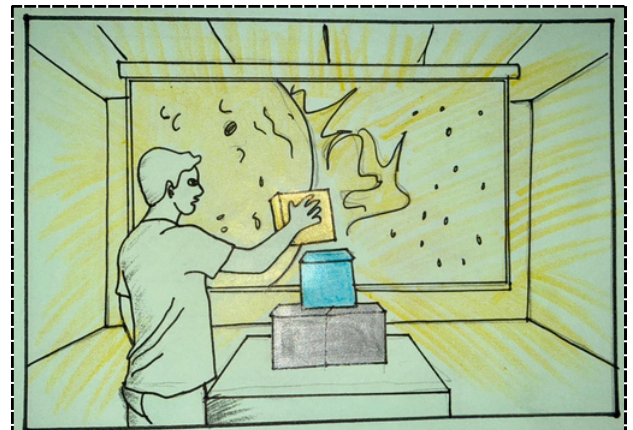
Frame 2: First Interaction

- Action: The user physically connects the Earth cube to the Moon cube.



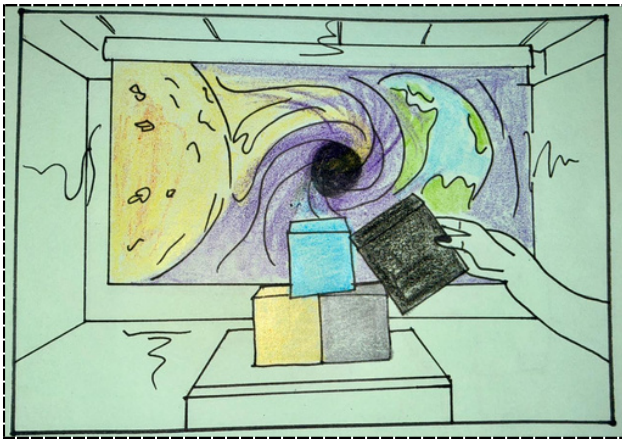
Frame 3: Stability

- Description: A soft blue glow appears, revealing a lifelike 3D Earth with its Moon gently orbiting. A smooth, steady sine wave hums in sync, echoing the calm rhythm between them.



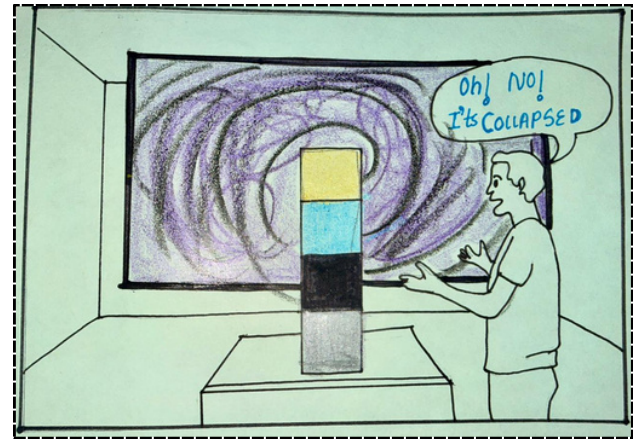
Frame 4: Balanced ecosystem

- Action: The user slides the star cube into the stack, touching the Earth & Moon pair.
- Description: The room fills with light, creates radiant solar flares and golden rays that sweep across the Earth. The sound expands into a warm, harmonic chord, reflecting a balanced solar system.



Frame 5: Tension

- Action: The user picks up the Black Hole cube and slowly brings it toward the existing stack.
- Description: As the cube moves closer, shifting resistance and voltage, the visuals start to jitter and distort. The Earth and Moon shake off their steady paths, while the sound glitches and crackles, building a sense of tension.



Frame 6: Collapse

- Action: The user places the Black Hole cube firmly onto the stack, completing the circuit.
- Description: The big whirlpool shows up, sucking in the light and planets. The sound gets louder and harsher until the screen turns purple, white then everything goes black.

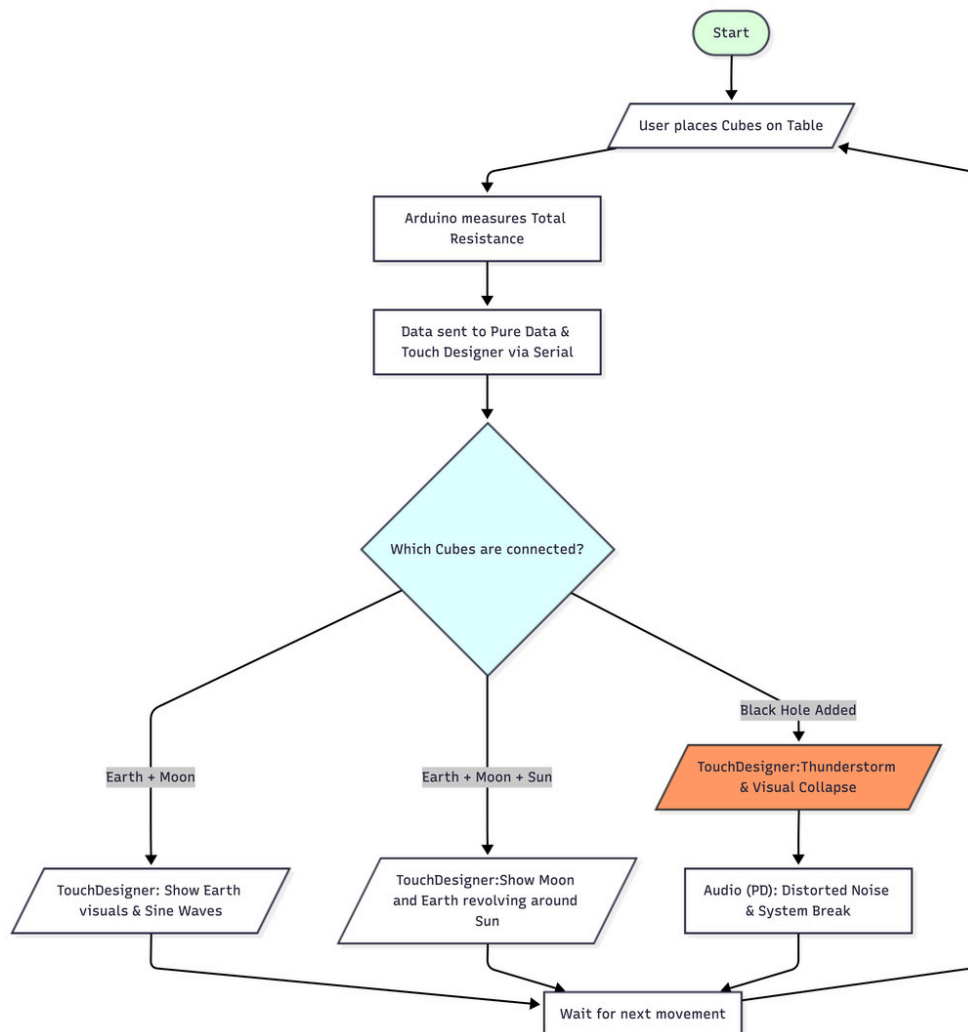
The storyboard was one of the first things the group created. Its job was to map out the emotional journey of the piece from emptiness, through life and connection, and finally to collapse. Before any of the technical work came together, the storyboard gave everyone a shared sense of how the experience should feel and where it was going dramatically.

As the project developed, the interaction model grew richer than the storyboard originally suggested. Rather than only responding to combinations, the system was built to recognise each individual cuboid on its own. This gave the audience something more exploratory they could discover what each celestial body looked and sounded like alone before starting to connect them.

The final interaction states, as shown in the presentation and video, were:

- star alone -> solar visuals and sound
- Earth alone -> planetary visuals and sound
- Moon alone -> lunar visuals and sound
- Black Hole alone -> dark, ominous visuals and sound
- star + Earth -> layered visuals and audio
- star + Earth + Moon -> full solar system, richest audiovisual state
- star + Earth + Moon + Black Hole -> collapse sequence, total destruction

The flowchart designed alongside the storyboard captured the system logic clearly: the user places cubes, the Arduino measures total resistance, that data is sent via Serial to TouchDesigner and Pure Data, the software identifies which combination is active, triggers the matching audiovisual response, and then loops back waiting for the next change.



RESEARCH & WIDER CONTEXT

What Inspired Us

The idea for this project grew out of a genuine fascination with space how stars are born, how solar systems form, and the stories people have told about the night sky for thousands of years. Space felt like the right subject because it belongs to everyone. In the most literal sense, it is universal.

We were also drawn to the fact that a solar system already has a story built into it: it begins, it grows, and eventually it ends. That arc of creation and collapse mapped naturally onto the modular structure of the assignment, where each new cuboid changes the whole system. The concept and the technology felt like they were saying the same thing.

Interactive Art We Looked At

Semiconductor – Ruth Jarman & Joe Gerhardt

Semiconductor is an audiovisual duo who build installations using real scientific data from space solar wind, electromagnetic signals and turn them into sound and image. What drew us to their work was the way they take something completely invisible and make it something you can actually feel. That was exactly what we were going for, just at a smaller and more hands-on scale. Their work also showed us that cosmic data could be beautiful without losing its scientific honesty.



Ruth Jarman and Joe Gerhardt
audiovisual installation



David Rokeby : Very Nervous System

David Rokeby – Very Nervous System (1986–1990)

David Rokeby's Very Nervous System is one of the most important works in the history of interactive art. Using cameras and synthesisers, he built a system that turned body movement directly into sound. What stayed with us was his belief that the body not a button or a screen should be the instrument. He wanted the interface to feel as invisible as possible, so that interaction felt natural rather than mechanical. Our project shares that ambition. The cuboids are meant to feel like objects worth picking up and handling, not like input devices. The interactivity should feel discovered, not operated.

Zimoun – Modular Sound Sculptures (ongoing)

Zimoun is a Swiss artist who builds large installations from simple, repeated components motors, wires, cardboard boxes that together create complex and evolving soundscapes. What connected his work to ours is the modular logic: one unit does one thing, but combine several and the system becomes something qualitatively different. That is exactly what we were building. His work also lives at the intersection of the physical and the sonic, which felt very close to our own approach.



Participatory Art



first ever photograph of
a black hole

Participatory Art

We also looked at participatory art more broadly work that only becomes complete when someone physically engages with it. This idea sat at the core of our project. Without an audience picking up the cuboids and placing them, nothing happens. People are not watching the solar system they are building it. And they are the ones who choose to end it.

The Black Hole as an Ending

We thought carefully about why the Black Hole felt like the right way to close the piece. In 2019, scientists released the first ever photograph of a black hole. In 2022, they captured the one at the centre of our own galaxy. Both felt like extraordinary moments something that had only ever existed in theory suddenly became real and visible. A black hole is not just destruction; it is also one of the most mysterious and compelling things in the known universe. Asking the audience to place it themselves knowing it will collapse everything they have built felt like an honest way to sit with that complexity.

TECHNICAL PROCESS

Individual Cuboid Construction

Each group member built one cuboid using a laser cutter, cutting panels from 3mm MDF sheet and assembling them into a 100mm cube. The outer faces were fitted with copper contact points, so that when two cuboids are pressed together, they conduct electricity between them.

Inside each cuboid, a network of resistors was wired across the faces and crucially, each cuboid was given its own unique resistor value. This means the system can identify not just how many cuboids are connected, but exactly which one is present at any moment. The Star, Earth, Moon, and Black Hole can each be recognised individually, so the right visuals and audio trigger no matter what order they are placed in.

Arduino – Reading the Sculpture

An Arduino Nano reads the voltage values produced by the resistor networks across four analog pins (A0–A3). The raw readings, which range from 0 to 1023, are normalised to a 0–1 floating-point range by dividing by 1023.0. This keeps the values consistent even when there are small fluctuations in voltage.

These four values are sent as a comma-separated string over serial communication at 9600 baud, with a 50ms delay between readings producing a continuous stream that looks something like: [0.36, 0.31, 0.27, 0.24].

Because each cuboid has a distinct resistor value, each one produces a distinct reading, giving TouchDesigner everything it needs to identify what is connected and respond accordingly.

Visuals and Sound in TouchDesigner

TouchDesigner was chosen as the main audiovisual environment because of how naturally it handles real-time, data-driven performance. A Serial DAT node receives the comma-separated values from the Arduino, parses them, and routes them into four channels that drive all the visual and audio behaviour in the patch. Seven distinct visual states were programmed in total four individual and three cumulative:

- Star alone — warm golden solar imagery, radiant flares and stellar light filling the projection
- Earth alone — blue-green planetary visuals, a living world suspended in dark space
- Moon alone — cool grey lunar imagery, the Moon as a solitary satellite in silence
- Black Hole alone — dark, swirling visuals, waiting in isolation
- Star + Earth — solar light and planetary visuals layering together on screen
- Star + Earth + Moon — the full solar system, visuals at their richest and most balanced
- Star + Earth + Moon + Black Hole — collapse sequence: light and planets swallowed until the image fades to white, then to total black

Audio Design

A single continuous music track plays throughout the entire experience Solar System Ambience | Space Sounds & Lo-fi Vibes, sourced from YouTube. Rather than switching tracks for each state, the music acts as a constant backdrop, an unbroken thread running from the void, through the building of the solar system, and into its collapse. What changes is not the music itself, but the audience's relationship to it.

The key audio moment is a volume dip. Every time a new cuboid is placed and the circuit changes, the music briefly lowers before returning to full volume. This was implemented in TouchDesigner by detecting the moment a channel value changes and triggering a short fade-down and fade-up on the audio output. It is subtle, but it matters it gives each placement a felt pause, a breath, as if the universe briefly holds still in response to being touched.

Working Touch designer File



Code in Arduino IDE

```
TouchDesignerCode [Arduino IDE 2.3.6]
File Edit Sketch Tools Help
Arduino Nano

TouchDesignerCode.ino
1 void setup() {
2   Serial.begin(9600);
3 }
4
5 void loop() {
6   float b1 = analogRead(A0) / 1023.0;
7   float b2 = analogRead(A1) / 1023.0;
8   float b3 = analogRead(A2) / 1023.0;
9   float b4 = analogRead(A3) / 1023.0;
10
11   // send everything in one line
12   Serial.print(b1);
13   Serial.print(",");
14   Serial.print(b2);
15   Serial.print(",");
16   Serial.print(b3);
17   Serial.print(",");
18   Serial.print(b4);
19   delay(50);
20 }
21 }
```

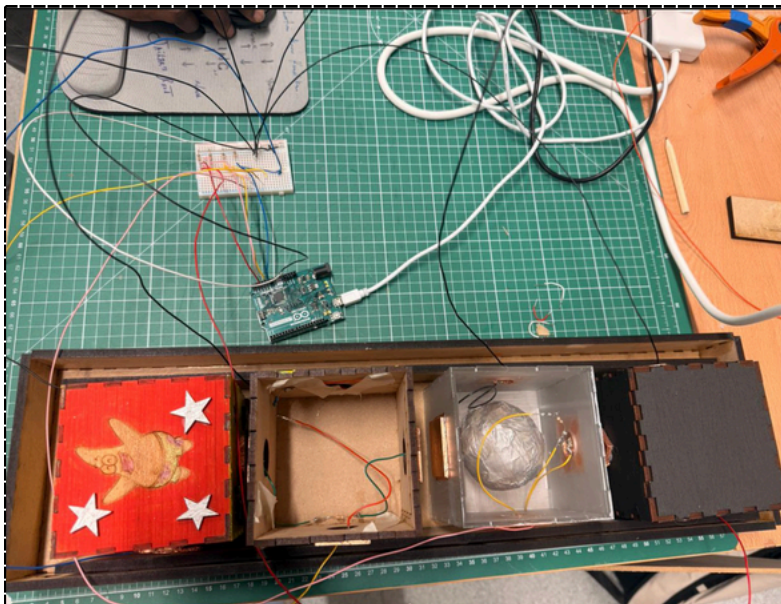
Output Serial Monitor X

Message (Enter to send message to 'Arduino Nano' on 'COM8')

New Line 9600 baud

```
0.26,0.31,0.27,0.27
0.29,0.29,0.26,0.26
0.29,0.29,0.28,0.27
0.26,0.26,0.26,0.26
0.26,0.26,0.26,0.26
0.26,0.26,0.26,0.26
```

The Working Connections



PROJECT RESULTS

The completed sculpture functioned as a fully interactive, real-time audiovisual installation. The following outcomes were achieved across the technical and experiential dimensions of the project.

Technical Outcomes

- Four individual laser-cut cuboids were constructed, each with a unique resistor value allowing the Arduino to identify each one independently.
- The Arduino Nano successfully read and normalised voltage values across four analog pins and transmitted them via Serial at 9600 baud.
- TouchDesigner received and parsed the Serial data in real time, correctly routing four channels to drive seven distinct audiovisual states.
- The collapse sequence - triggered by placing all four cuboids including the Black Hole - executed reliably, with the whirlpool visual and audio distortion culminating in a full blackout.
- The volume dip interaction functioned consistently throughout testing and the final presentation.

Experiential Outcomes

- Audience members engaged with the sculpture without instruction, discovering the interactive logic organically - confirming the design goal of immediate accessibility.
- The cosmic narrative arc from void to collapse was legible to viewers with no prior knowledge of the project concept.
- The piece successfully created a sense of agency and consequence: audiences understood that their physical choices were directly shaping what appeared on screen.

CONCEPTUAL REFLECTIONS

Working through this project made us think a lot more carefully about what interactivity actually means. It's easy to assume that if something responds to input, it's interactive but what we found is that the response needs to feel meaningful, not just technically functional. The hardest part was taking raw voltage numbers from the Arduino and turning them into something an audience would actually care about.

The volume dip on each placement is one of the decisions we're most happy with. It wasn't in the original plan it came out of testing. Without it, placing a cube felt like nothing. With it, every placement had a moment of stillness, like something was genuinely happening in the room. It made the physical action feel connected to what was appearing on screen.

The decision to make the Black Hole the final, destructive element was something we talked about a lot as a group. It means the experience ends on a dark note the audience destroys what they built. But we felt that was more interesting and more honest than wrapping things up positively. The whole point is that the audience makes that choice themselves. That felt too important to change.



TECHNICAL REFLECTIONS

The biggest problem we ran into was getting the cuboids to steady control the switching behaviour in TouchDesigner. This became the central issue on presentation day.

The signal from the Arduino was arriving correctly we could see the channel values updating inside the patch but the switch logic that was supposed to trigger the right visual state was not firing reliably. At one point we had to bypass the cuboids entirely and switch states manually by typing values into the patch. That was the opposite of what the project was meant to be.

The root cause was in the Switch node configuration. The incoming values needed to be correctly mapped before they could drive the switch, and this had not been tested under real conditions beforehand. Once we adjusted the channel expressions in TouchDesigner, the cuboids worked cleanly as intended.

This taught us the difference between a system that works in parts and one that works as a whole. Every component had been tested individually, but we had not done enough end-to-end testing before the day. In future, we would set aside dedicated time for full integration testing well in advance. A secondary issue was the copper contact points between cuboids slight misalignment could cause readings to fluctuate. A future version might use magnetic or gold-plated connectors for a more stable connection.

Despite these challenges, the system worked during the final presentation. The moment the cuboids took control of the screen was one of the most satisfying points of the whole project.